What's Wrong With Science?

When Should We Believe Science?

Publication · p-hacking · Replication · Fraud

Session 5 · Professor David Gresham · Fall 2026

Guest: Ivan Oransky, co-founder of Retraction Watch

Image: Ioannidis (2005), PLOS Medicine, CC BY

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Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

Summary

There is increasing concern that most current published research findings are false. The probability that a research claim is true may depend on study power and bias, the number of other studies on the same question, and, importantly, the ratio of true to no relationships among the relationships probed in each scientific field. In this framework, a research finding is less likely to be true when the studies conducted in a field are smaller; when effect sizes are smaller; when there is a greater number and lesser preselection of tested relationships; where there is greater flexibility in designs, definitions, outcomes, and analytical modes; when there is greater financial and other interest and prejudice; and when more teams are involved in a scientific field in chase of statistical significance. Simulations show that for most study designs and settings, it is more likely for a research claim to be false than true. Moreover, for many current scientific fields, claimed research findings may often be simply accurate measures of the prevailing bias. In this essay, I discuss the implications of these problems for the conduct and interpretation of research.

Published research findings are sometimes refuted by subsequent evidence, with ensuing confusion and disappointment. Refutation and controversy is seen across the range of research designs, from clinical trials and traditional epidemiological studies [1–3] to the most modern molecular research [4,5]. There is increasing concern that in modern research, false findings may be the majority or even the vast majority of published research claims [6–8]. However, this should not be surprising. It can be proven that most claimed research findings are false. Here I will examine the key factors that influence this problem and some corollaries thereof.

The Essay section contains opinion pieces on topics of broad interest to a general medical audience.

Modeling the Framework for False Positive Findings

Several methodologists have pointed out [9–11] that the high rate of nonreplication (lack of confirmation) of research discoveries is a consequence of the convenient, yet ill-founded strategy of claiming conclusive research findings solely on the basis of a single study assessed by formal statistical significance, typically for a *p*-value less than 0.05. Research is not most appropriately represented and summarized by *p*-values, but, unfortunately, there is a widespread notion that medical research articles

It can be proven that most claimed research findings are false.

should be interpreted based only on *p*-values. Research findings are defined here as any relationship reaching formal statistical significance, e.g., effective interventions, informative predictors, risk factors, or associations. "Negative" research is also very useful. "Negative" is actually a misnomer, and the misinterpretation is widespread. However, here we will target relationships that investigators claim exist, rather than null findings.

As has been shown previously, the probability that a research finding is indeed true depends on the prior probability of it being true (before doing the study), the statistical power of the study, and the level of statistical significance [10,11]. Consider a 2×2 table in which research findings are compared against the gold standard of true relationships in a scientific field. In a research field both true and false hypotheses can be made about the presence of relationships. Let R be the ratio of the number of "true relationships" to "no relationships" among those tested in the field. R is characteristic of the field and can vary a lot depending on whether the field targets highly likely relationships or searches for only one or a few true relationships among thousands and millions of hypotheses that may be postulated. Let us also consider, for computational simplicity, circumscribed fields where either there is only one true relationship (among many that can be hypothesized) or the power is similar to find any of the several existing true relationships. The pre-study probability of a relationship being true is $R/(R + 1)$. The probability of a study finding a true relationship reflects the power $1 - \beta$ (one minus the Type II error rate). The probability of claiming a relationship when none truly exists reflects the Type I error rate, α. Assuming that c relationships are being probed in the field, the expected values of the 2×2 table are given in Table 1. After a research finding has been claimed based on achieving formal statistical significance, the post-study probability that it is true is the positive predictive value, PPV. The PPV is also the complementary probability of what Wacholder et al. have called the false positive report probability [10]. According to the 2×2 table, one gets $PPV = (1 - \beta)R/(R - \beta R + \alpha)$. A research finding is thus

Citation: Ioannidis JPA (2005) Why most published research findings are false. PLoS Med 2(8): e124.

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Abbreviation: PPV, positive predictive value

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Competing Interests: The author has declared that no competing interests exist.

DOI: 10.1371/journal.pmed.0020124



Ivan Oransky

Co-founder, Retraction Watch · Editor-in-Chief, The Transmitter · Distinguished Journalist in Residence, NYU



? / 100

If 100 studies published in top journals were repeated exactly, how many would get the same result?

If 100 studies published in top journals were repeated exactly, how many would get the same result?

(A) 0-25%

0%

(B) 25-50%

0%

(C) 50 -75%

0%

(D) 75 -100%

0%

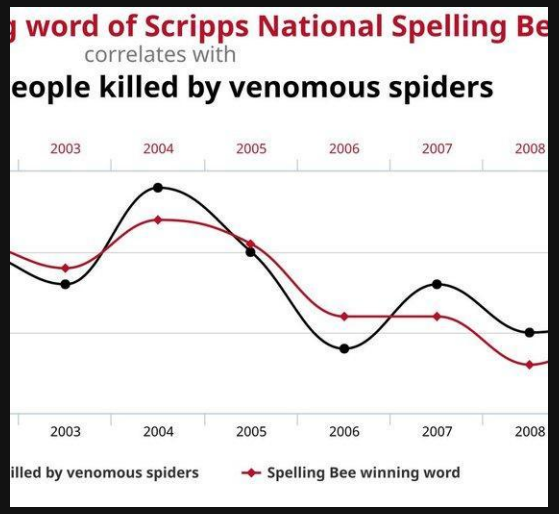
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Four questions for tonight



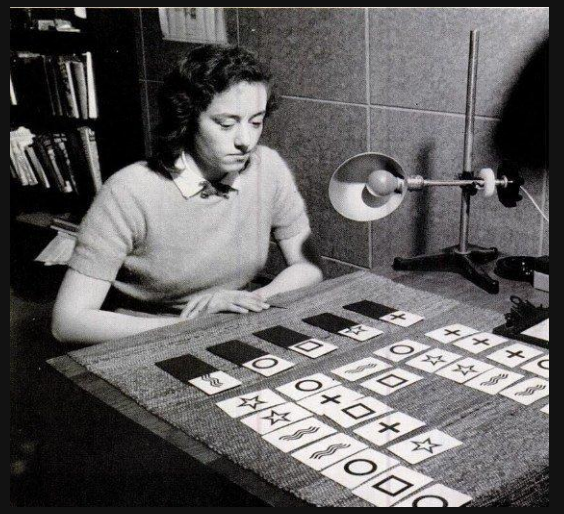
1 Publication

What science gets published?



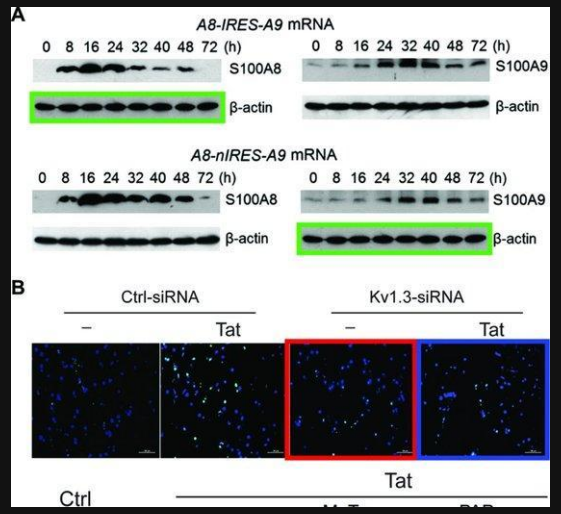
2 p-hacking

How do honest scientists find things that aren't there?



3 Replication

How much of it holds up?



4 Fraud

What happens when it isn't honest?

Two studies. Same question. Which gets published?

JOURNAL OF EVERYDAY SCIENCE

Daily coffee cuts depression risk by 40%

p = 0.01

A

Two studies. Same question. Which gets published?

JOURNAL OF EVERYDAY SCIENCE

Daily coffee cuts depression risk by 40%

p = 0.01

A

JOURNAL OF EVERYDAY SCIENCE

Daily coffee has no effect on depression

p = 0.62

B



PUBLICATION BIAS

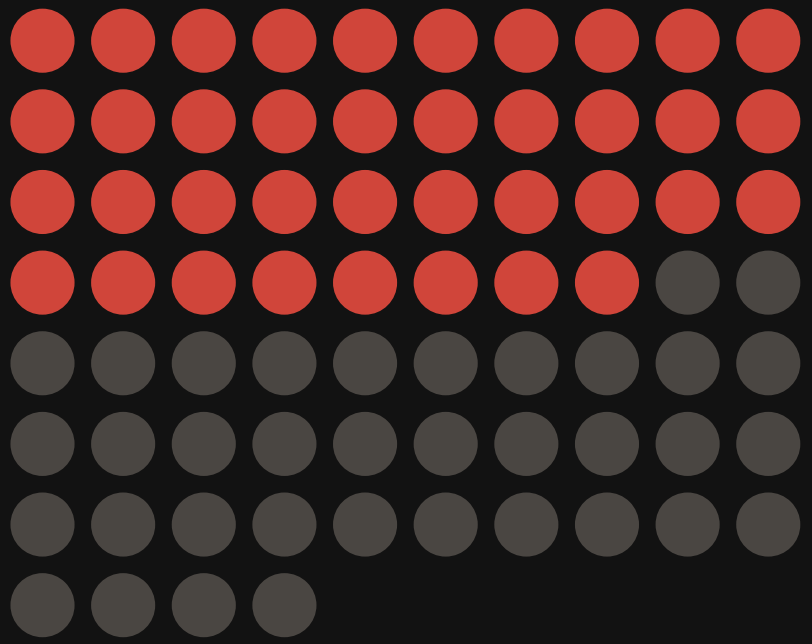
The file drawer

Positive results get published.
Null results get filed away.

“Selective Publication of Antidepressant Trials and Its Influence on Apparent Efficacy.” *The New England Journal of Medicine* 358 (3): 252–60. Turner et al., 2008

Turner et al., New England Journal of Medicine, 2008

“Selective Publication of Antidepressant Trials and Its Influence on Apparent Efficacy.” *The New England Journal of Medicine* 358 (3): 252–60. Turner et al., 2008



51%

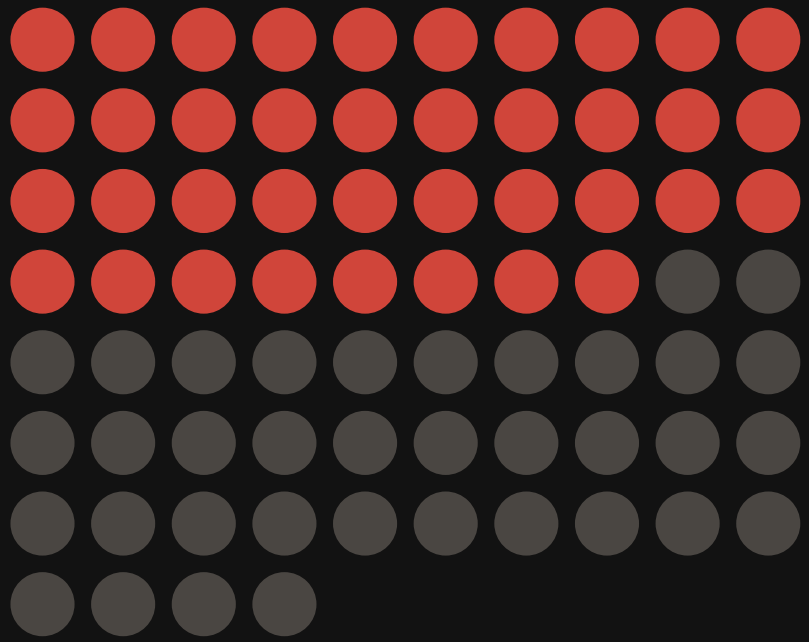
positive

All 74 trials submitted to the FDA

12 antidepressants · 12,564 patients

Turner et al., New England Journal of Medicine, 2008

“Selective Publication of Antidepressant Trials and Its Influence on Apparent Efficacy.” *The New England Journal of Medicine* 358 (3): 252–60. Turner et al., 2008



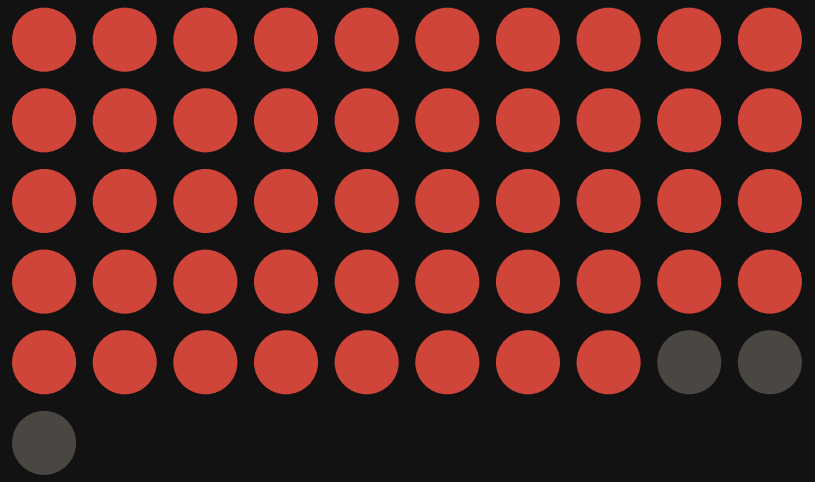
51%

positive

All 74 trials submitted to the FDA

12 antidepressants · 12,564 patients

38 positive results
36 negative or questionable



94%

positive

The 51 trials published in journals

22 negative trials never published; 11 published as if positive
3 published as negative results

Turner et al., New England Journal of Medicine, 2008



2011 · THE REPLICATION CRISIS BEGINS

Feeling the future

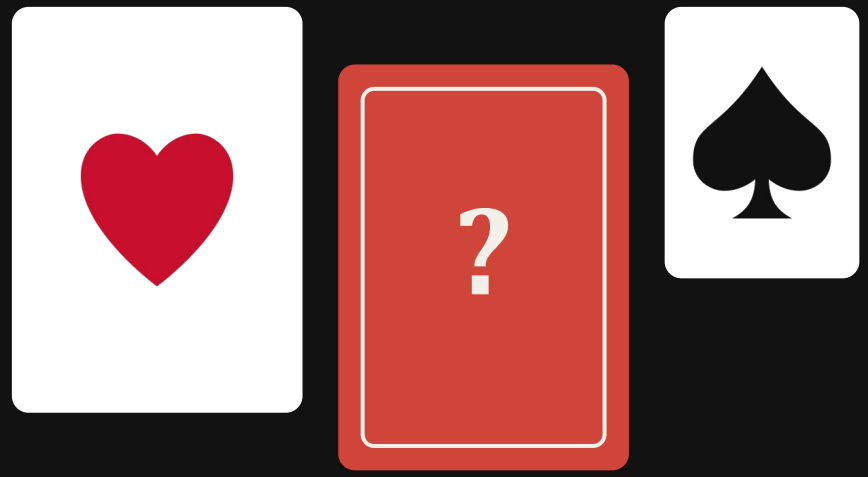
A top psychology journal published 9 experiments, with over 1,000 participants, showing people can sense events before they happen.

Either ESP is real, or the standard methods are broken.

Daryl Bem, Journal of Personality and Social Psychology, 2011. Photo: ESP card test, 1930s.

ACTIVITY · ARE YOU PSYCHIC?

Tonight we test this room for ESP



- Write down the numbers 1–10.
- For each card, write
  - R (red) or
  - B (black)
    - before I turn it over.
- Give yourself a
  - checkmark if you get it right
  - a cross if you guess wrong
- Add up your total score

How many did you get right?

(A) 0 — 0%

(B) 1 — 0%

(C) 2 — 0%

(D) 3 — 0%

(E) 4 — 0%

(F) 5 — 0%

(G) 6 — 0%

(H) 7 — 0%

(I) 8 — 0

(J) 9 — 0

SEE MORE

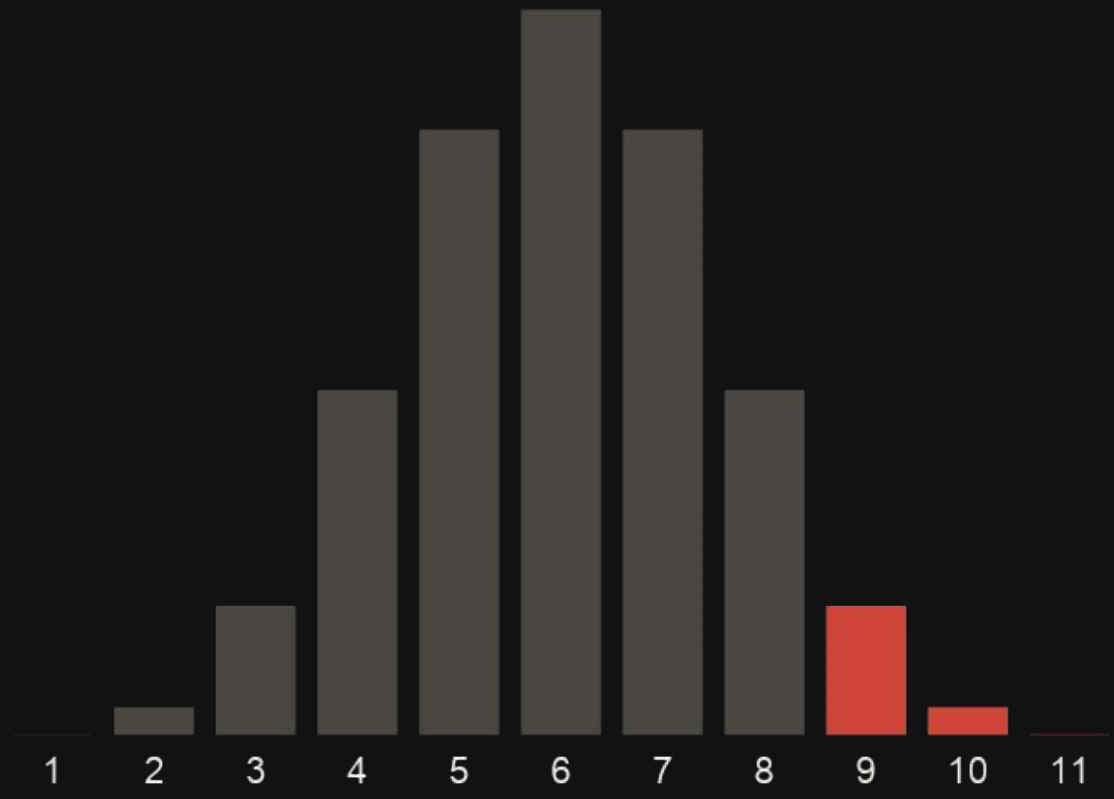
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9 or more?

Stand up.

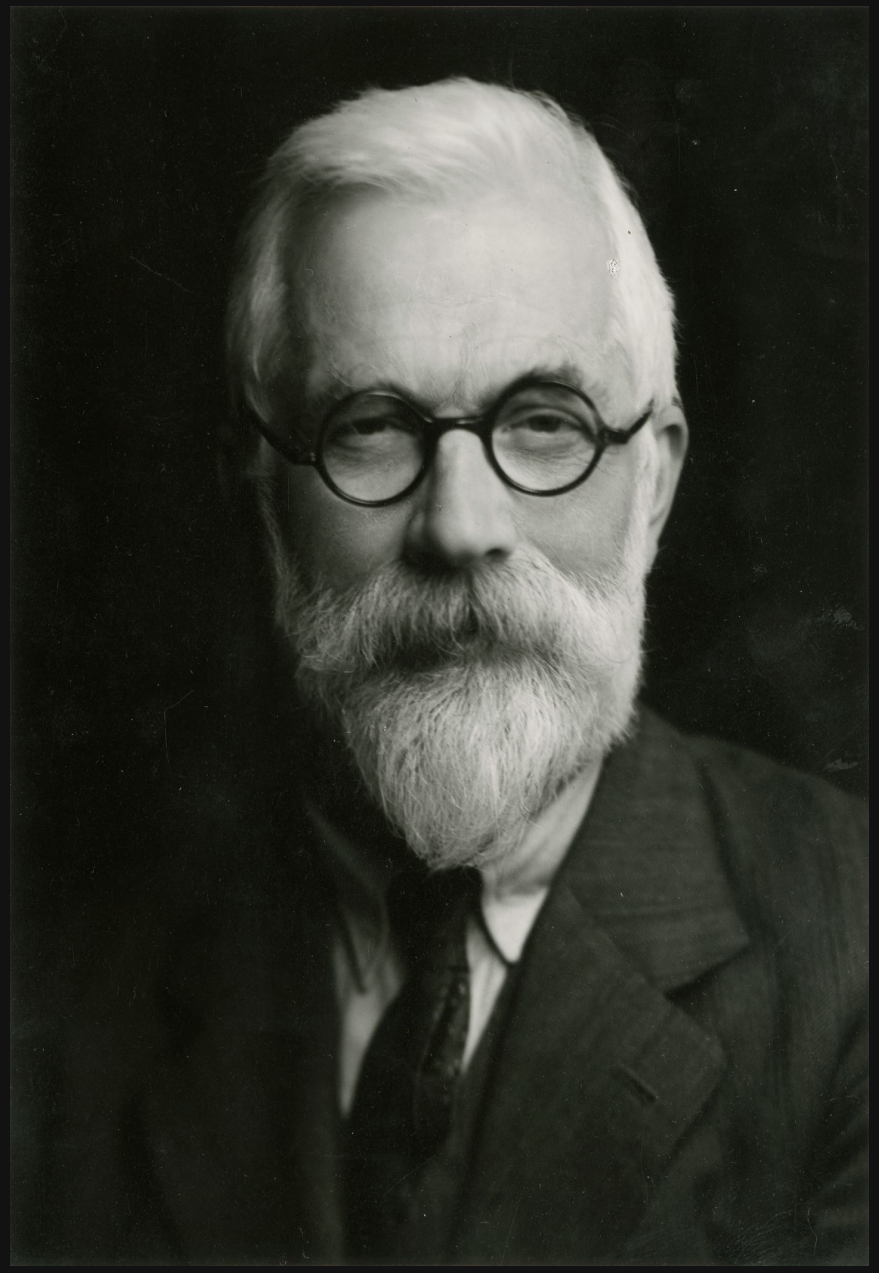
$p < 0.05$. You're published in the Journal of Psychic Research.

Everyone else: you're the file drawer.



What just happened?

Nobody cheated. So where did the “discovery” come from?



$p < 0.05$

If there were no real effect, a result this striking would turn up less than 1 time in 20.

It does not mean a 95% chance the finding is true.

Ronald Fisher, who called 1 in 20 a "convenient" cut-off (1925)

Test 20 things that don’t work…



…and expect one “discovery.”

Data Dredging



p-hacking

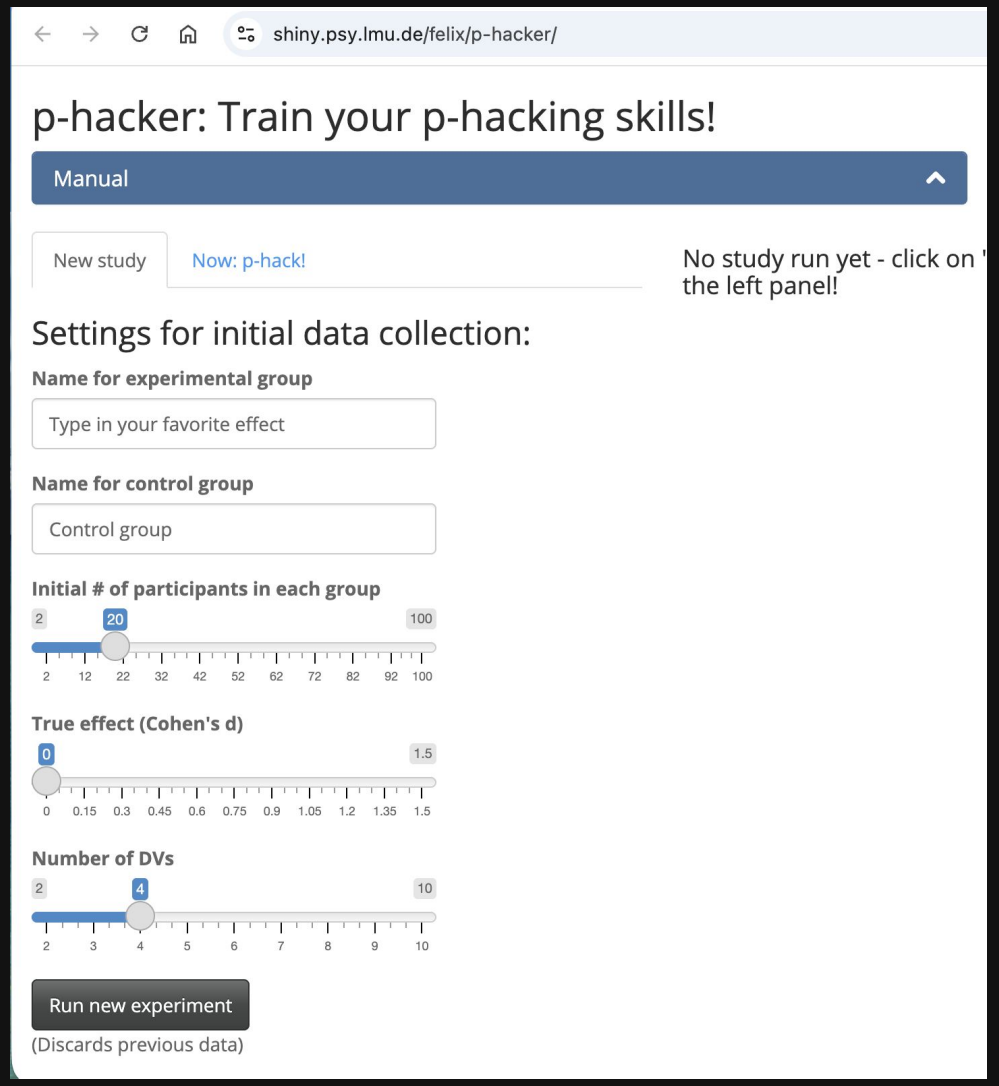
the misuse of data analysis to force a statistically significant result (a *p*-value below 0.05), even when no real effect exists.

p-hacking

the misuse of data analysis to force a statistically significant result (a *p*-value below 0.05), even when no real effect exists.

- **Optional stopping:** Running an experiment and stopping data collection the moment a *p*-value drops below 0.05.
- **Selective reporting:** Testing dozens of variables or outcomes, but only publishing the one or two that are significant by chance.
- **Data trimming:** Excluding outliers, removing specific participants, or adding covariates after looking at the data to force the numbers into alignment

https://shiny.psy.lmu.de/felix/p-hacker/



How to prevent p-hacking

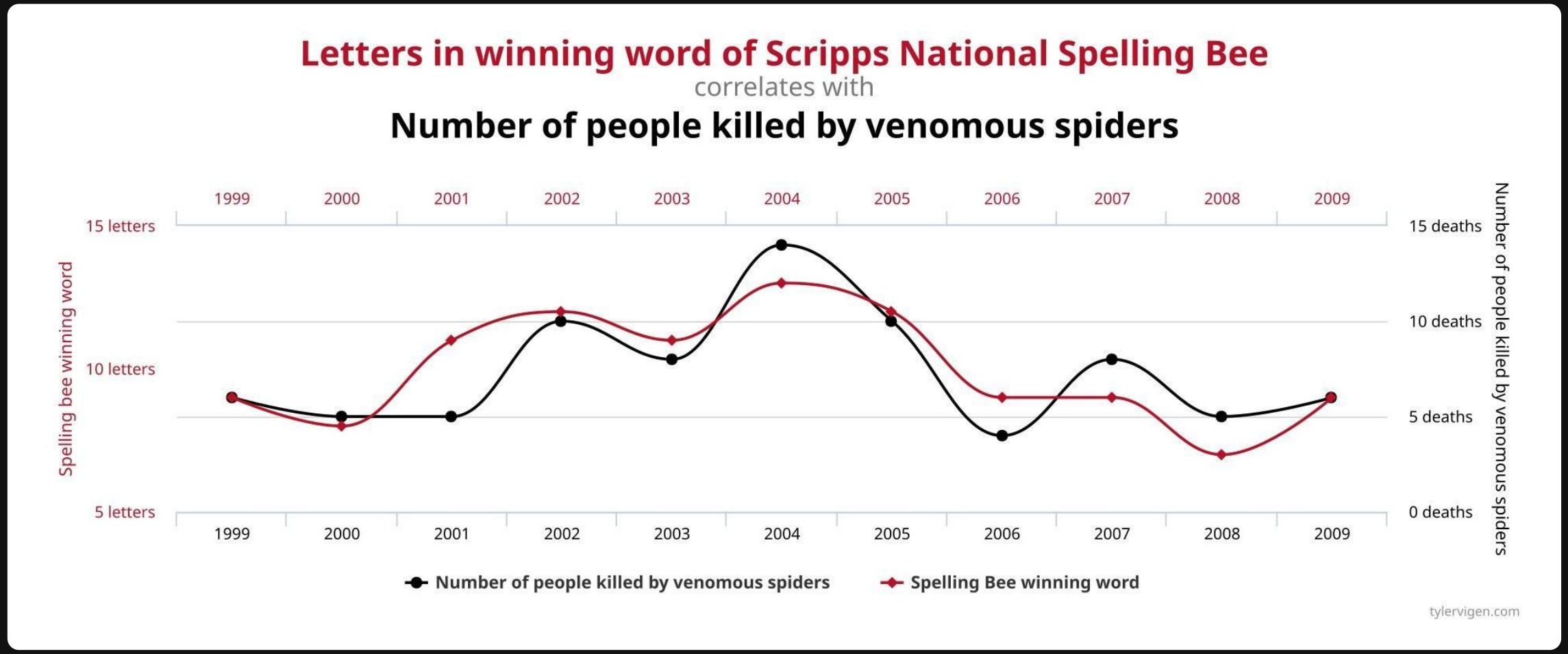
- **Preregistration**: Writing down your hypothesis, sample size, and analysis plan before looking at any data.
- **Transparency**: Sharing raw data and reporting all tests conducted, not just the successful ones.
- S**tatistical corrections**: Using methods like the Bonferroni Correction to adjust significance thresholds when running multiple tests.

HARKing
(Hypothesis after results are known)



Making up a new hypothesis *after* seeing the results so it looks like it was predicted all along.

With enough data, something always correlates



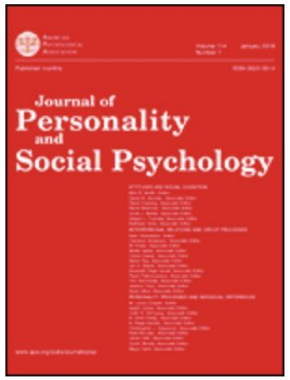
Spurious Correlations https://tylervigen.com/

English Logi

APA PsycNet®

AMERICAN PSYCHOLOGICAL ASSOCIATION

SEARCH BROWSE Recent Searches



Journal Information
Journal TOC

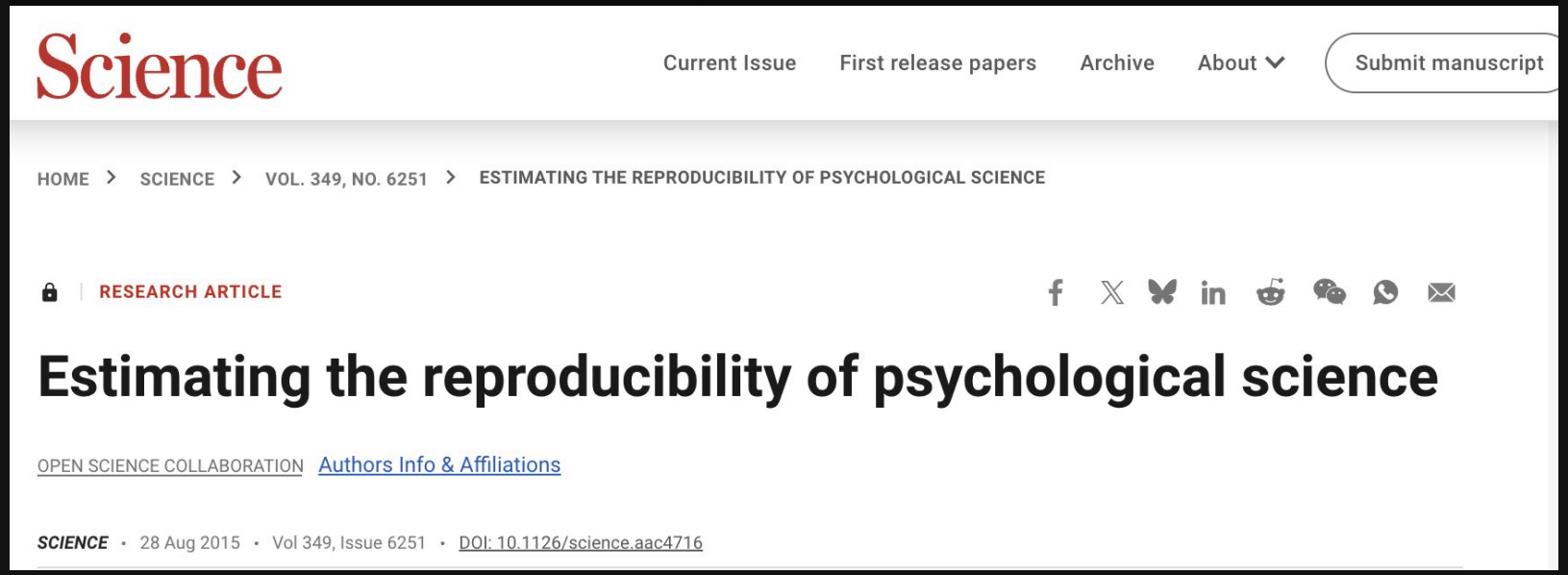
APA PsycArticles: Journal Article

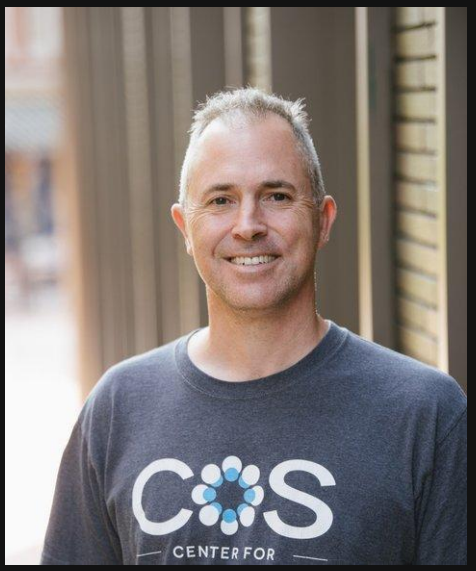
Feeling the future: Experimental evidence for anomalous retroactive influences on cognition and affect.

Bem, D. J. (2011). Feeling the future: Experimental evidence for anomalous retroactive influences on cognition and affect. *Journal of Personality and Social Psychology, 100*(3), 407–425. https://doi.org/10.1037/a0021524

The term psi denotes anomalous processes of information or energy transfer that are currently unexplained in terms of known physical or biological mechanisms. Two variants of psi are *precognition* (conscious cognitive awareness) and premonition (affective apprehension) of a future event that could not otherwise be anticipated through any known inferential process. Precognition and *premonition* are themselves special cases of a more general phenomenon: the anomalous retroactive influence of some future event on an individual's current responses, whether those responses are conscious or nonconscious, cognitive or affective. This article reports 9 experiments, involving more than 1,000 participants, that test for retroactive influence by “time-reversing” well-established psychological effects so that the individual's responses are obtained before the putatively causal stimulus events occur. Data are presented for 4 time-reversed effects: precognitive approach to erotic stimuli and precognitive avoidance of negative stimuli; retroactive priming; retroactive habituation; and retroactive facilitation of recall. The mean effect size (d) in psi performance across all 9 experiments was 0.22, and all but one of the experiments yielded statistically significant results. The individual-difference variable of stimulus seeking, a component of extraversion, was significantly correlated with psi performance in 5 of the experiments, with participants who scored above the midpoint on a scale of stimulus seeking achieving a mean effect size of 0.43. Skepticism about psi, issues of replication, and theories of psi are also discussed. (PsycInfo Database Record (c) 2025 APA, all rights reserved)

How many psychology studies can be replicated?



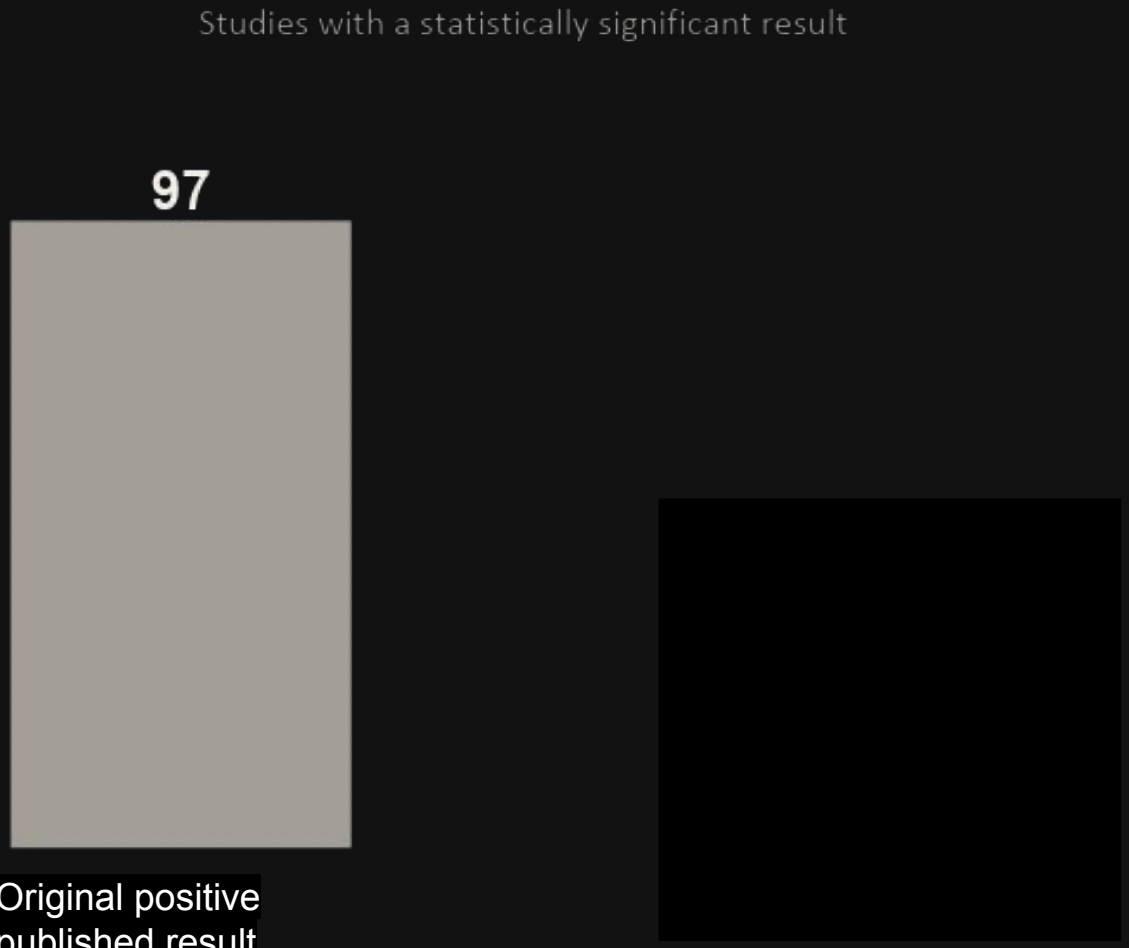


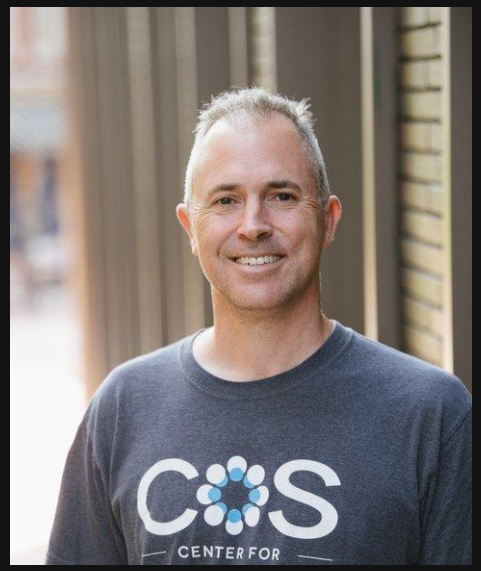
270 researchers repeated 100 published psychology studies

Led by Brian Nosek, Center for Open Science.

Open Science Collaboration, Science, 2015

How many psychology studies can be replicated?



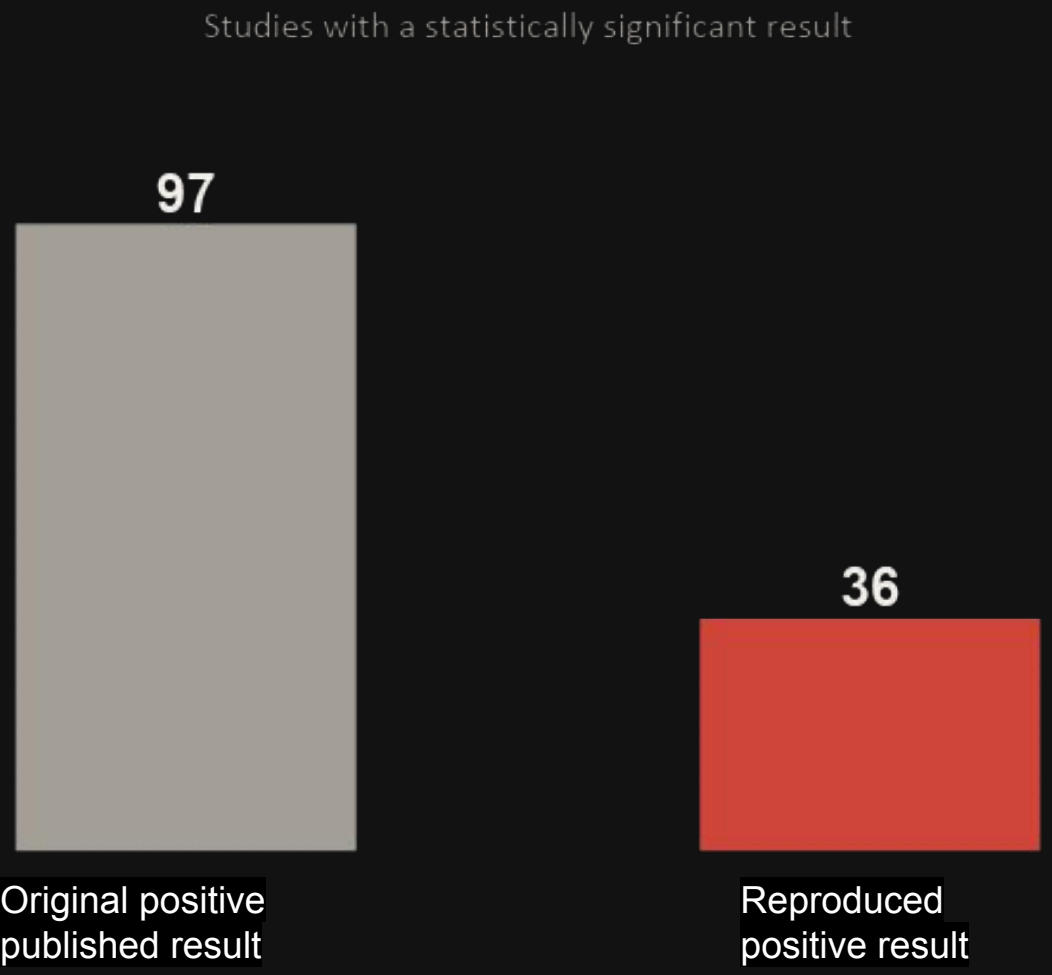


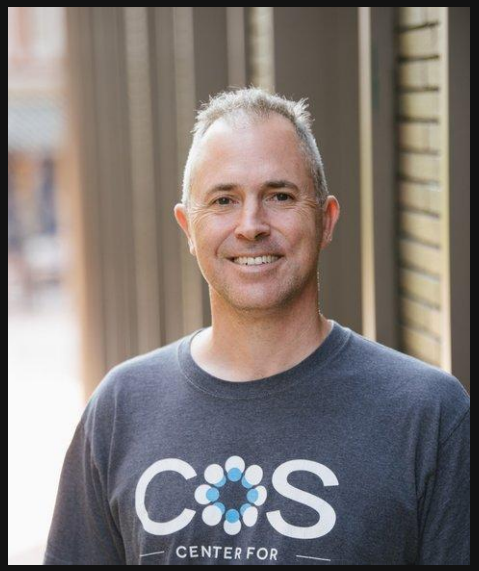
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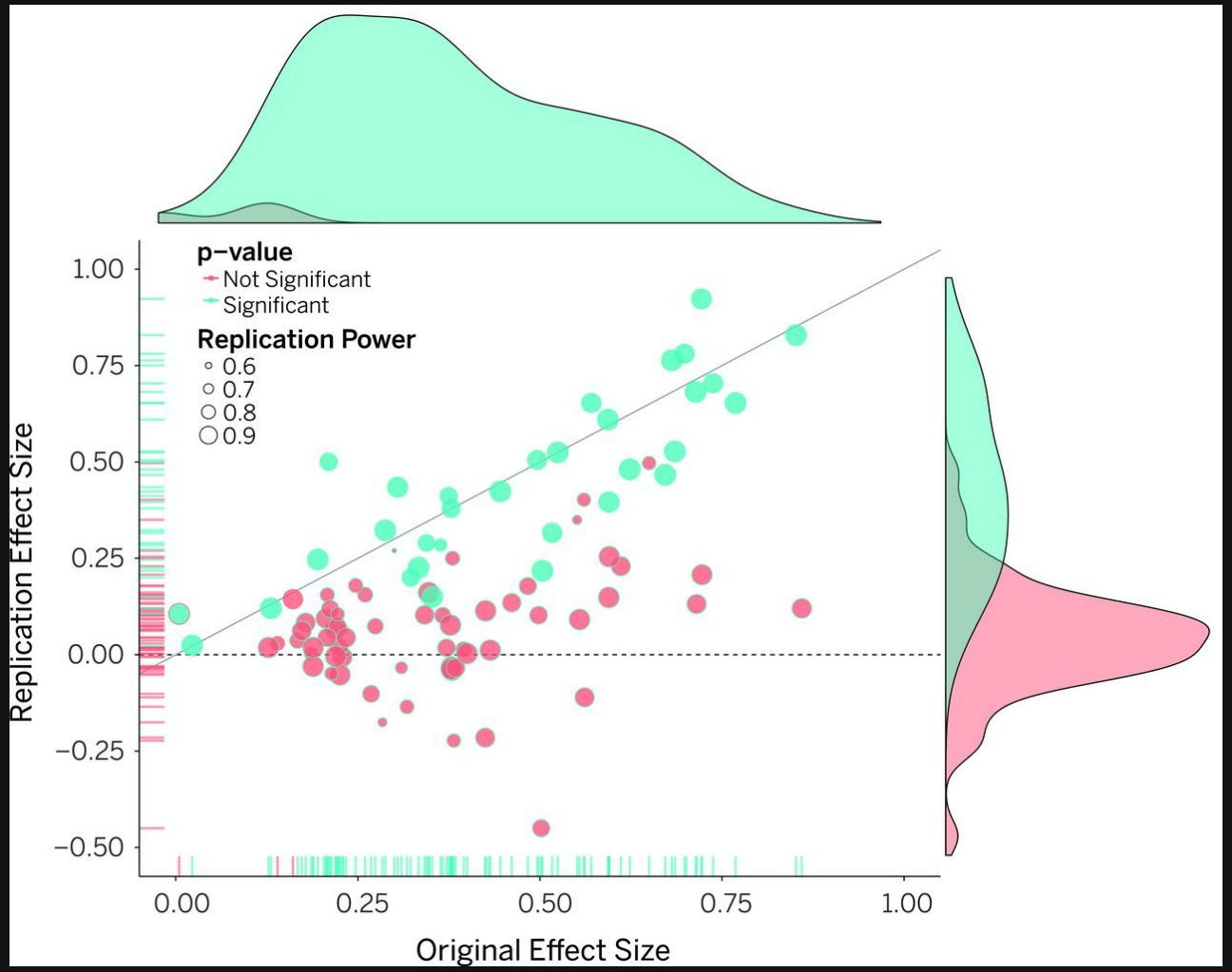


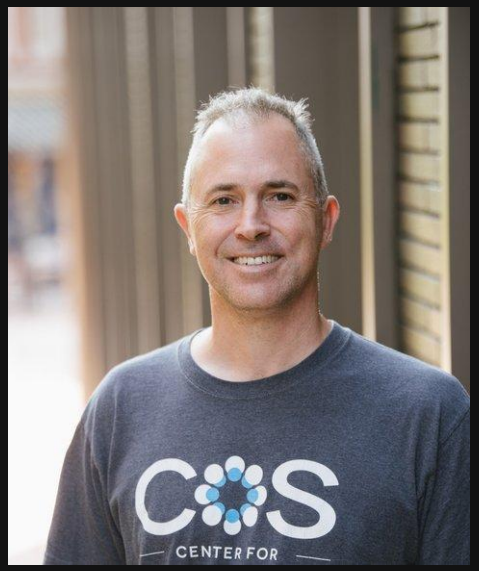
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Open Science Collaboration, Science, 2015

How many psychology studies can be replicated?





270 researchers repeated 100 published psychology studies

Led by Brian Nosek, Center for Open Science.

Open Science Collaboration, Science, 2015

Many cancer pre-clinical studies cannot be replicated

Raise standards for preclinical cancer research

C. Glenn Begley and **Lee M. Ellis** propose how methods, publications and incentives must change if patients are to benefit.

6 of 53

landmark cancer-biology findings confirmed when Amgen tried to repeat them (2012)

the demand for more effective therapies.

Over the past decade, before pursuing a particular line of research, scientists (including C.G.B.) in the haematology and oncology department at the biotechnology firm Amgen in Thousand Oaks, California, tried to confirm published findings related to that work. Fifty-three papers were deemed 'landmark' studies (see 'Reproducibility of research findings'). It was acknowledged from the outset that some of the data might not hold up, because papers were deliberately selected that described something completely new, such as fresh approaches to targeting cancers or alternative clinical uses for existing therapeutics. Nevertheless, scientific findings were confirmed in only 6 (11%) cases. Even knowing the limitations of preclinical research, this was a shocking result.

Many cancer pre-clinical studies cannot be replicated

Research Article

Cancer Biology, Computational and Systems Biology

Investigating the replicability of preclinical cancer biology

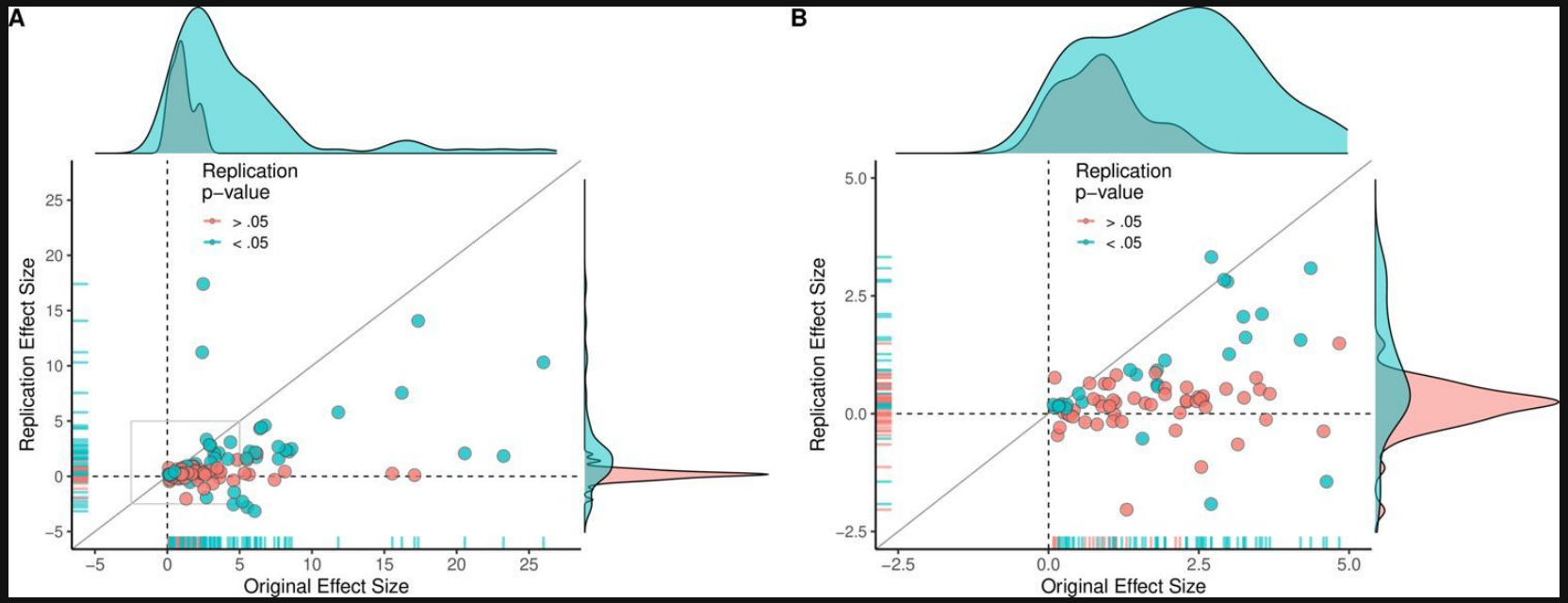
Timothy M Errington, Maya Mathur, Courtney K Soderberg, Alexandria Denis, Nicole Perfito, Elizabeth Iorns, Brian A Nosek

Center for Open Science, United States; Quantitative Sciences Unit, Stanford University, United States; Science Exchange, United States; University of Virginia, United States

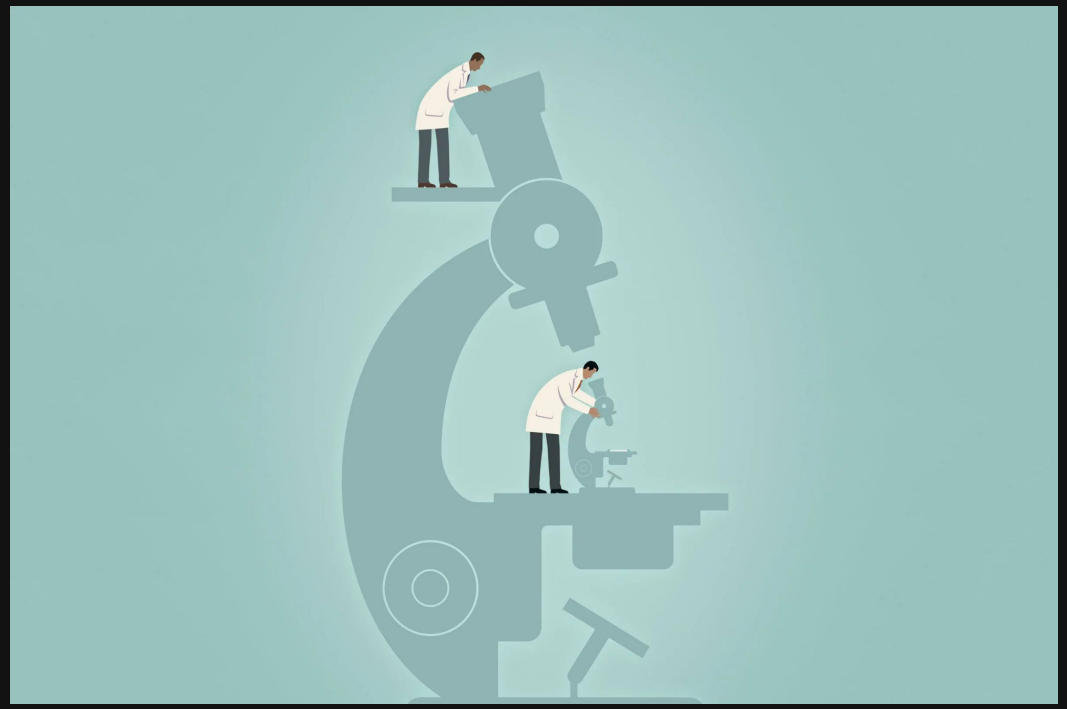
Dec 7, 2021 • https://doi.org/10.7554/eLife.71601

Either

- **No Positive effect**
- **Smaller effect size**



The Replication Crisis



From honest error to fraud



Honest error

A mislabeled sample, a coding bug



Questionable practices

p-hacking, HARKing, the file drawer.



Fabrication

Inventing or falsifying data

Most of the replication crisis lives in the middle.

Sloppiness of Deceit?

Spot the duplication

This figure passed peer review. Some panels appear more than once, labeled as different experiments.

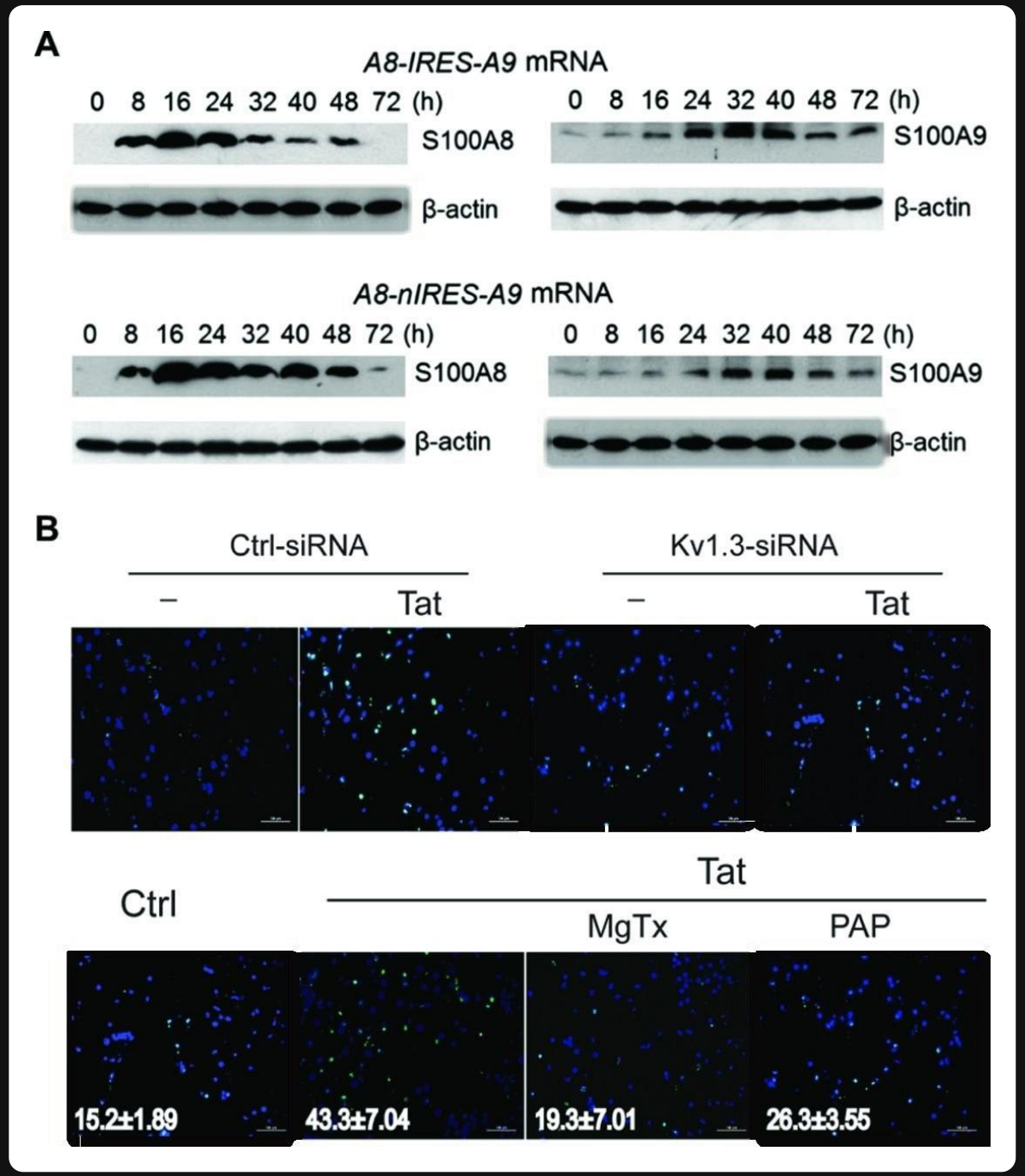


Figure: Bik, Casadevall & Fang, mBio, 2016, CC BY 4.0 (annotation boxes removed for this exercise).

Same β-actin blot (control)

used as the loading control for two different experiments

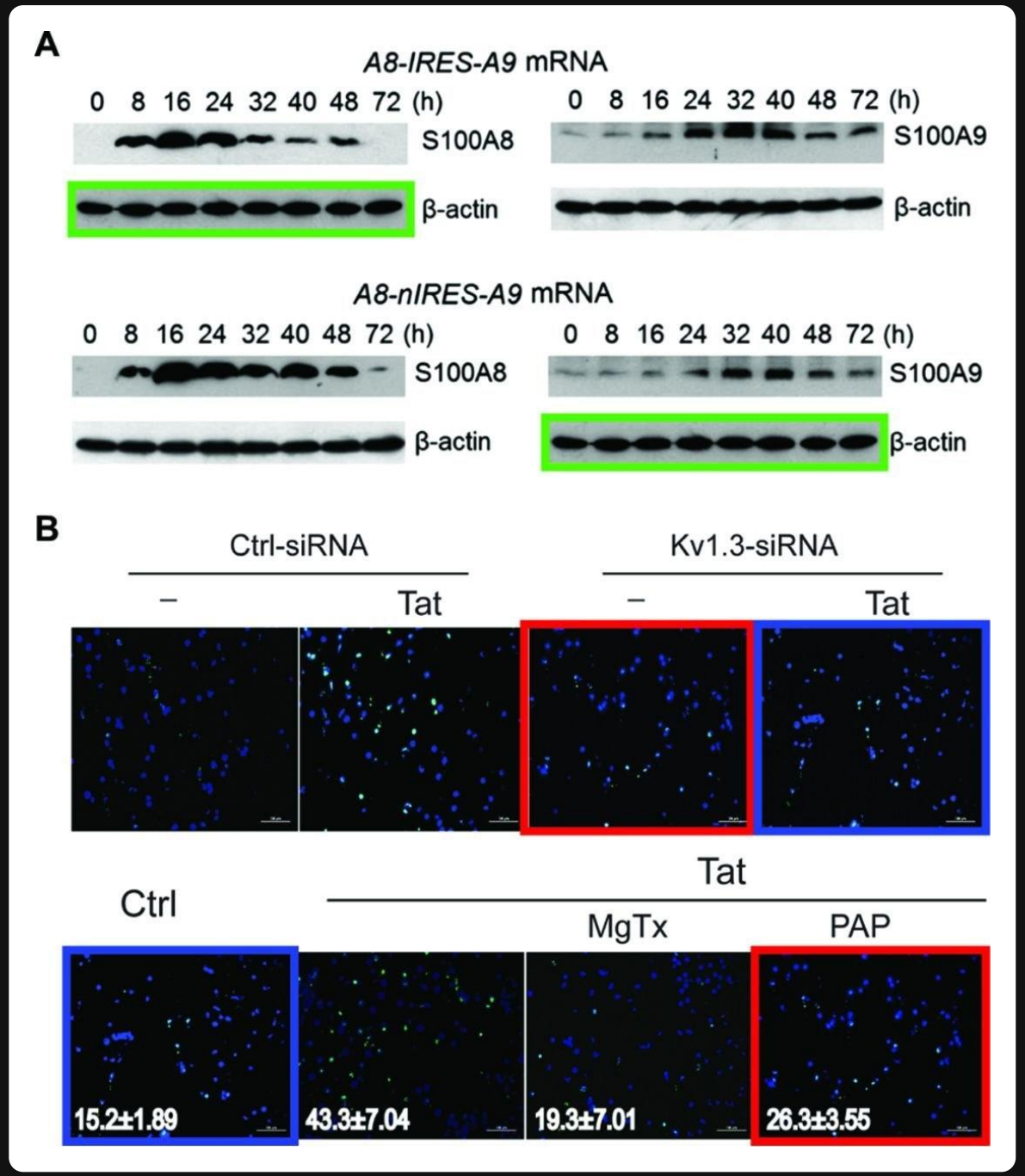
Same micrograph

labeled “Kv1.3-siRNA, no Tat” and “Tat + PAP”

Same micrograph

labeled “Kv1.3-siRNA + Tat” and “Ctrl”

Passed peer review…



Bik, Casadevall & Fang, “The Prevalence of Inappropriate Image Duplication in Biomedical Research Publications,” mBio, 2016, CC BY 4.0



Who catches it? Mostly outsiders.

Elisabeth Bik

Microbiologist turned full-time image sleuth

~4%

of 20,621 biomedical papers she screened contained problematic duplicated images

Also: Data Colada, PubPeer, journalists, junior colleagues. Rarely peer review.

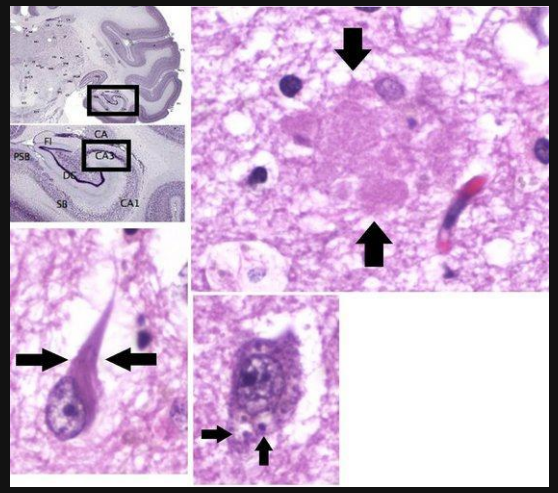
When it isn't honest



Andrew Wakefield

The Lancet, 1998

Vaccines and autism, 12 children. Retracted 12 years later.



Alzheimer's plaques

Nature, 2006

Doctored images in a landmark paper. Retracted 2024.



The honesty study

PNAS, 2012

Signing at the top reduces cheating. Data were fabricated. Retracted 9 years later

> Lancet. 1998 Feb 28;351(9103):637-41. doi: 10.1016/s0140-6736(97)11096-0.

Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children

A J Wakefield [1], S H Murch, A Anthony, J Linnell, D M Casson, M Malik, M Berelowitz, A P Dhillon, M A Thomson, P Harvey, A Valentine, S E Davies, J A Walker-Smith

Affiliations + expand

PMID: 9500320 DOI: 10.1016/s0140-6736(97)11096-0

Abstract

Background: We investigated a consecutive series of children with chronic enterocolitis and regressive developmental disorder.

Methods: 12 children (mean age 6 years [range 3-10], 11 boys) were referred to a paediatric gastroenterology unit with a history of normal development followed by loss of acquired skills, including language, together with diarrhoea and abdominal pain. Children underwent gastroenterological, neurological, and developmental assessment and review of developmental records. Ileocolonoscopy and biopsy sampling, magnetic-resonance imaging (MRI), electroencephalography (EEG), and lumbar puncture were done under sedation. Barium follow-through radiography was done where possible. Biochemical, haematological, and immunological profiles were examined.

Findings: Onset of behavioural symptoms was associated, by the parents, with measles, mumps, and rubella vaccination in eight of the 12 children, with measles infection in one child, and otitis media in another. All 12 children had intestinal abnormalities, ranging from lymphoid nodular hyperplasia to aphthoid ulceration. Histology showed patchy chronic inflammation in the colon in 11 children and reactive ileal lymphoid hyperplasia in seven, but no granulomas. Behavioural disorders included autism (nine), disintegrative psychosis (one), and possible postviral or vaccinal encephalitis (two). There were no focal neurological abnormalities and MRI and EEG tests were normal. Abnormal laboratory results were significantly raised urinary methylmalonic acid compared with age-matched controls ($p=0.003$), low haemoglobin in four children, and a low serum IgA in four children.

Interpretation: We identified associated gastrointestinal disease and developmental regression in a group of previously normal children, which was generally associated in time with possible environmental triggers.

EARLY REPORT · Volume 351, Issue 9103, P637-641, February 28, 1998 Download Full Issue



RETRACTED: Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children

Dr AJ Wakefield, FRCS [a] · SH Murch, MB [b] · A Anthony, MB [a] · J Linnell, PhD [a] · DM Casson, MRCP [b] · M Malik, MRCP [b] · et al.

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Summary

Background

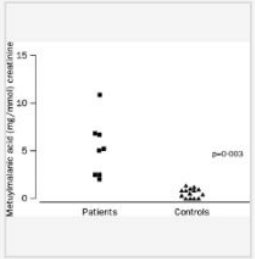
We investigated a consecutive series of children with chronic enterocolitis and regressive developmental disorder.

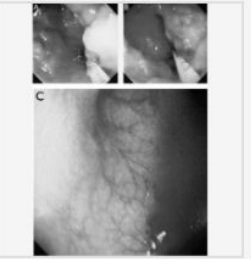
Methods

12 children (mean age 6 years [range 3–10], 11 boys) were referred to a paediatric gastroenterology unit with a history of

RETRACTED

Figures (3) Figure Viewer





Reasons for the retraction include:

- **Data Fabrication:** Andrew Wakefield altered medical histories and fabricated diagnoses for the 12 children in the study, falsely claiming a connection between the MMR vaccine, bowel disease, and autism.
- **Undisclosed Conflicts of Interest:** Andrew Wakefield received hundreds of thousands of pounds from lawyers suing vaccine manufacturers and held a financial interest in a competing single-measles vaccine patent, neither of which he disclosed.
- **Unethical Conduct:** The study was conducted without proper ethical approval and involved invasive, unnecessary medical procedures on children, such as lumbar punctures and colonoscopies

The tip of the iceberg?

4–9

retractions per 10,000 papers at most major publishers



Retraction Watch

Tracking retractions as a window into the scientific process

Top 10 most highly cited retracted papers

Ever curious which retracted papers have been most cited by other scientists? Below, we present the list of the 10 most highly cited retractions as of July 27, 2026. Readers will see some familiar entries, such as the in-

Retractions by Nobel Prize winners

In 2017, our co-founders wrote an intentionally ridiculous column in *Slate*: "Want to Win a Nobel Prize? Retract a Paper." But Nobel Prize winners have indeed retracted papers — some before and some after their awards. Here's our list.



1. Gregg Semenza (15): 2019 Nobel Prize in Physiology or Medicine. See coverage.
2. David Baltimore (3): 1975 Nobel Prize in Physiology or Medicine. See coverage.
3. Linda B Buck (3): 2004 Nobel Prize in Physiology or Medicine. See

IN CONVERSATION

Ivan Oransky

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Does science correct itself?

Retraction Watch

